

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

|                     |                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------|
| <b>Product Name</b> | <b><u>3-(Methylthio)propionaldehyde</u></b>                                                        |
| <b>CAS No</b>       | 3268-49-3                                                                                          |
| <b>Synonyms</b>     | 3-(Methylmercapto)propionaldehyde; Beta-(Methylmercapto)propionaldehyde; Methional; 4-Thiapentanal |

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>L08187</b>                                                                                        |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

No hazards identified

#### Health hazards

|                                    |            |
|------------------------------------|------------|
| Acute Oral Toxicity                | Category 4 |
| Acute Dermal Toxicity              | Category 3 |
| Acute Inhalation Toxicity - Vapors | Category 4 |
| Skin Corrosion/Irritation          | Category 2 |
| Serious Eye Damage/Eye Irritation  | Category 1 |
| Skin Sensitization                 | Category 1 |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Chronic aquatic toxicity | Category 3 |
|--------------------------|------------|

#### Label Elements



Skull and Crossbones



Corrosion

**Signal Word****Danger****Hazard Statements**

H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H311 - Toxic in contact with skin  
H318 - Causes serious eye damage  
H412 - Harmful to aquatic life with long lasting effects  
H302 + H332 - Harmful if swallowed or if inhaled  
Combustible liquid

**Precautionary Statements**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse  
P403 + P235 - Store in a well-ventilated place. Keep cool  
P405 - Store locked up  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Stench  
Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component      | CAS No    | Weight % |
|----------------|-----------|----------|
| 4-Thiapentanal | 3268-49-3 | >95      |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Self-Protection of the First Aider</b>  | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Causes eye burns. May cause allergic skin reaction. Causes severe eye damage. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Sulfur oxides, Thermal decomposition can lead to release of irritating gases and vapors.

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition. Risk of ignition. Thermal decomposition can lead to release of irritating gases and vapors. In the event of fire and/or explosion do not breathe fumes. Combustible material. Containers may explode when heated.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

**Clean-up methods - small spillage**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |
| Neoprene       |                                   |                 |                 |                       |
| Natural rubber |                                   |                 |                 |                       |
| PVC            |                                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g.

sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices                                                                                        |
| <b>Recommended Filter type:</b>        | Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                                                                                                                                         |
| <b>Recommended half mask:-</b>         | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                                                                                                                |
| <b>Hygiene Measures</b>                | When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing. Keep away from food, drink and animal feeding stuffs. Contaminated work clothing should not be allowed out of the workplace. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wear suitable gloves and eye/face protection. |
| <b>Environmental exposure controls</b> | Do not allow material to contaminate ground water system. Prevent product from entering drains.                                                                                                                                                                                                                                                                                                                    |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                         |                                        |
|------------------------------------------------|-----------------------------------------|----------------------------------------|
| <b>Appearance</b>                              | Colorless to pale yellow                |                                        |
| <b>Physical State</b>                          | Liquid                                  |                                        |
| <b>Odor</b>                                    | Stench                                  |                                        |
| <b>Odor Threshold</b>                          | No data available                       |                                        |
| <b>pH</b>                                      | No information available                |                                        |
| <b>Melting Point/Range</b>                     | -75 °C / -103 °F                        |                                        |
| <b>Softening Point</b>                         | No data available                       |                                        |
| <b>Boiling Point/Range</b>                     | 170.3 °C / 338.5 - 338.5 °F             | @ 760 mm Hg                            |
| <b>Flash Point</b>                             | 69 °C / 156.2 °F                        | <b>Method -</b> CC (closed cup)        |
| <b>Evaporation Rate</b>                        | No data available                       |                                        |
| <b>Flammability (solid,gas)</b>                | Not applicable                          | Liquid                                 |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.3%<br><b>Upper</b> 26.1% |                                        |
| <b>Vapor Pressure</b>                          | 23 hPa @ 20 °C                          |                                        |
| <b>Vapor Density</b>                           | 3.59                                    | (Air = 1.0)                            |
| <b>Specific Gravity / Density</b>              | 1.050                                   |                                        |
| <b>Bulk Density</b>                            | Not applicable                          | Liquid                                 |
| <b>Water Solubility</b>                        | No information available                |                                        |
| <b>Solubility in other solvents</b>            | No information available                |                                        |
| <b>Partition Coefficient (n-octanol/water)</b> |                                         |                                        |
| <b>Component</b>                               | <b>log Pow</b>                          |                                        |
| 4-Thiapentanal                                 | 0.34                                    |                                        |
| <b>Autoignition Temperature</b>                | No data available                       |                                        |
| <b>Decomposition Temperature</b>               | No data available                       |                                        |
| <b>Viscosity</b>                               | No data available                       |                                        |
| <b>Explosive Properties</b>                    |                                         | explosive air/vapour mixtures possible |
| <b>Oxidizing Properties</b>                    | No information available                |                                        |
| <b>Other information</b>                       |                                         |                                        |
| <b>Molecular Formula</b>                       | C4 H8 O S                               |                                        |
| <b>Molecular Weight</b>                        | 104.17                                  |                                        |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                                                                                        |
| <b>Stability</b>                        | Stable under normal conditions.                                                                                                                   |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition.                                             |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Bases.                                                                                                                   |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Sulfur oxides. Thermal decomposition can lead to release of irritating gases and vapors. |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                          |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                            |            |
|----------------------------|------------|
| <b>(a) acute toxicity;</b> |            |
| <b>Oral</b>                | Category 4 |
| <b>Dermal</b>              | Category 3 |
| <b>Inhalation</b>          | Category 4 |

| Component      | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation              |
|----------------|--------------------------|------------------------------|------------------------------|
| 4-Thiapentanal | LD50 = 496 mg/kg ( Rat ) | LD50 = 1700 mg/kg ( Rabbit ) | LC50 > 4.84 mg/L ( Rat ) 4 h |

|                                       |            |
|---------------------------------------|------------|
| <b>(b) skin corrosion/irritation;</b> | Category 2 |
|---------------------------------------|------------|

|                                           |            |
|-------------------------------------------|------------|
| <b>(c) serious eye damage/irritation;</b> | Category 1 |
|-------------------------------------------|------------|

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b>(d) respiratory or skin sensitization;</b> |                          |
| <b>Respiratory</b>                            | No data available        |
| <b>Skin</b>                                   | Category 1               |
| <b>Sensitization</b>                          | No information available |

|                                    |                   |
|------------------------------------|-------------------|
| <b>(e) germ cell mutagenicity;</b> | No data available |
|------------------------------------|-------------------|

|                             |                                                           |
|-----------------------------|-----------------------------------------------------------|
| <b>(f) carcinogenicity;</b> | No data available                                         |
|                             | There are no known carcinogenic chemicals in this product |

|                                   |                   |
|-----------------------------------|-------------------|
| <b>(g) reproductive toxicity;</b> | No data available |
|-----------------------------------|-------------------|

|                                  |                   |
|----------------------------------|-------------------|
| <b>(h) STOT-single exposure;</b> | No data available |
|----------------------------------|-------------------|

|                                    |                           |
|------------------------------------|---------------------------|
| <b>(i) STOT-repeated exposure;</b> | No data available         |
| <b>Target Organs</b>               | No information available. |

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## Section 12 - Ecological Information

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Contains a substance which is: Toxic to aquatic organisms.

| Component      | Freshwater Fish | Water Flea | Freshwater Algae                                    | Microtox |
|----------------|-----------------|------------|-----------------------------------------------------|----------|
| 4-Thiapentanal |                 |            | EC50: = 2.45 mg/L, 72h<br>(Desmodesmus subspicatus) |          |

**Persistence and Degradability** No information available  
**Persistence** Persistence is unlikely.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.  
**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component      | log Pow | Bioconcentration factor (BCF) |
|----------------|---------|-------------------------------|
| 4-Thiapentanal | 0.34    | No data available             |

**Mobility** .  
**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

## Section 14 - Transport Information

### IMDG/IMO

UN-No UN2785  
Proper Shipping Name 4-THIAPENTANAL  
Hazard Class 6.1  
Packing Group III

### ADG

UN-No UN2785  
Proper Shipping Name 4-THIAPENTANAL  
Hazard Class 6.1

|                                     |                     |
|-------------------------------------|---------------------|
| <b>Packing Group</b>                | III                 |
| <b>Component</b>                    | <b>Hazchem Code</b> |
| 4-Thiapentanal<br>3268-49-3 ( >95 ) | 2X                  |

**IATA**

**UN-No** UN2785  
**Proper Shipping Name** 4-THIAPENTANAL  
**Hazard Class** 6.1  
**Packing Group** III

**Environmental hazards** No hazards identified

**Special Precautions** No special precautions required

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                  | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|----------------------------|-------------------------------------------------------------|------------------------|
| 4-Thiapentanal - 3268-49-3 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### **International Inventories**



| Component      | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL |
|----------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|------|
| 4-Thiapentanal | X    | X     | 221-882-5 | -      | X    | X   | -    | X     | X    | X    | X     | -    |

**Legend:** X - Listed, '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**  
Not applicable.

| Component      | CAS No    | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------|-----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 4-Thiapentanal | 3268-49-3 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH** Not applicable

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**NZS 5433:2012** - Transport of Dangerous Goods on Land  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**PNEC** - Predicted No Effect Concentration  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail  
**OECD** - Organisation for Economic Co-operation and Development  
**LC50** - Lethal Concentration 50%  
**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
 Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date**

18-Nov-2022

**Revision Summary**

Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**